Chen-Chun Chen

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Education

The Ohio State University **2020–Present**
PhD, Computer Science and Engineering
- Advisor: Dhabaleswar K. Panda
- Dissertation: Designing GPU-Aware Collective Communication for Heterogeneous Clusters with Diverse GPUs and Interconnects

National Tsing Hua University **2016–2018**
MS, Computer Science
- Advisor: Jerry Chou
- Thesis: ALBERT: an Automatic Learning Based Execution and Resource Management System for Hadoop

National Tsing Hua University **2012–2016**
BS, Computer Science

Summary

PhD candidate in Computer Science and Engineering specializing in designing, implementing, and optimizing GPU-aware MPI libraries and scalable collective algorithms for heterogeneous HPC systems. Experienced with NVIDIA, AMD, and Intel GPUs, with 20+ peer-reviewed publications contributing to MVAPICH. Skilled in bridging low-level communication and middleware to build portable, efficient, and high-performance solutions for large-scale scientific computing and AI applications.

Work Experience

The Ohio State University (The Network-Based Computing Laboratory)
Graduate Research Associate *Columbus, OH* *Aug 2020–Present*
- Contribute to the MVAPICH project, focusing on GPU-aware MPI runtimes and performance optimization.
- Design and implement scalable collective communication algorithms (Allreduce, Alltoall).
- Collaborate with research institutions to evaluate designs on leading supercomputers.
- Publish and present research on HPC communication runtimes.

X-ScaleSolutions, LLC
Software Engineer Intern *Columbus, OH* *May 2022–Aug 2022*
- Ran and profiled PETSc workloads on CPU and GPU systems.
- Identified performance bottlenecks and optimized execution efficiency.

Taiwan Semiconductor Manufacturing Company (TSMC)
Engineer *Hsinchu, Taiwan* *Sep 2018–Jul 2019*
- Designed and maintained automation scripts and programs for production line operations.
- Automated production line operations with scripts and deployment tools, improving workflow efficiency by 20%.
- Built and deployed a cluster health-checking system, detected and prevented at least 3 anomalies in 2 months.
- Conducted research using deep learning to predict production yield based on transistor density.

Taiwan Semiconductor Manufacturing Company (TSMC)
Intern Engineer *Hsinchu, Taiwan* *Jul 2017–Aug 2017*
- Applied deep learning to predict job execution times in live production environments.
- Designed job scheduling and optimization strategies to improve throughput with minimal overhead.

The Ohio State University, National Tsing Hua University
Graduate Teaching Associate
- OSU: Algorithm (AU 2020)
- NTHU: Cloud Programming (SP 2017, SP 2018), Parallel Programming (AU 2016, AU 2017), Machine Learning (AU 2017)

Current Projects

GPU-Aware MPI [5, 6, 8, 9, 10, 11, 12, 14, 18]
- Extended MVAPICH to support GPU-aware communication.
- **IPC-based Alltoall Algorithm**: Designed hybrid Alltoall algorithms leveraging GPU zero-copy IPC, boosting heFFTe throughput by up to 28× on 128 GPUs compared to MVAPICH2-GDR.
- **Kernel-based Direct Allreduce Algorithm**: Achieved up to 23% lower latency than NCCL on large-scale systems and 40% faster performance than existing MPI libraries, resulting in 27% faster Amber simulations.
- **Intel GPU Support**: Developed GPU-aware MPI runtimes for point-to-point and collective operations, achieving 11× faster Allreduce on a single node, 42% improvement on 32 GPUs, and up to 20× higher Alltoallv throughput over MPICH.
- **Multi-rail-aware Allreduce Designs**: Delivered 2.8–2.9× speedup at 1 GB over Cray MPICH and Intel MPI by leveraging persistent GPU buffers, multi-leader communication, and pipelined designs across NVIDIA, AMD, and Intel GPUs.
- **Compression Framework for MPI Collectives**: Integrated casting-based compression into MPI collectives, reducing Allreduce latency by 42% over MVAPICH-Plus + ZFP and Alltoall latency by 59%.
- **MPI-xCCL**: Developed a portable MPI layer over vendor collective libraries (NCCL, RCCL, HCCL, MSCCL), enabling cross-accelerator portability with competitive performance without code changes.

Past Projects

Hybrid Compression Optimization for Distributed Training [1]

- Designed a performance-driven hybrid compression framework to accelerate distributed TensorFlow training with Horovod.

ALBERT: Automatic Learning-Based Execution and Resource Management for Hadoop [4]

- Integrated research results from the following 2 programs and extended them to a cloud platform using OpenStack Sahara.
- Applied machine learning, particularly deep learning, to predict Hadoop job running times in cloud environments.
- Designed an execution optimization mechanism that improved system throughput using a modified 2D bin-packing algorithm guided by time prediction results.

Job Prediction on Big-data platforms (with Institute for Information Industry (III), Taiwan) [21]

- Developed machine learning models to predict job running times on Hadoop and Spark platforms.
- Integrated existing modules in III to implement an automatic resource reallocate mechanism driven by prediction results.

Job Prediction and Resource Management on Production Line (with TSMC, Taiwan)

- Utilized deep learning techniques to forecast job execution times in real semiconductor production environments.
- Designed optimization strategies that improved total system throughput by leveraging prediction results without introducing additional overhead.

Student Cluster Competition (SCC) in HPC

- Competed in SC15 (2nd place overall), TSCC (Champion), and ASC16 (First Class Award).
- Tuned performance for HPC apps (WRF, LAMMPS) and ported neural networks to Intel Xeon Phi.

Languages

- C/C++
- Bash
- Python
- Java
- JavaScript
- Perl

Libraries/Frameworks

- MPI
- OpenMP
- ROCm/HIP
- CUDA
- oneAPI/SYCL
- Level Zero

Journals

[1] **Chen, C.-C.**, Chou, Y.-M., Chou, J., "PHY: A performance-driven hybrid communication compression method for distributed training". In: *Journal of Parallel and Distributed Computing* (2023).

[2] Suresh, K., Khorassani, K., **Chen, C.-C.**, Ramesh, B., Abduljabbar, M., Shafi, A., Subramoni, H., Panda, D. K., "Network-Assisted Noncontiguous Transfers for GPU-Aware MPI Libraries". In: *IEEE Micro* (May 2023).

[3] Khorassani, K. S., **Chen, C.-C.**, Ramesh, B., Shafi, A., Subramoni, H., Panda, D. K., "High Performance MPI over the Slingshot Interconnect". In: *Journal of Computer Science and Technology* (Feb. 2023).

[4] **Chen, C.-C.**, Wang, K.-S., Hsiao, Y.-T., Chou, J., "ALBERT: An automatic learning based execution and resource management system for optimizing Hadoop workload in clouds". In: *Journal of Parallel and Distributed Computing* (2022).

Conferences and Workshops

[5] **Chen, C.-C.**, Yao, J., Panda, D. K., "Design and Implementation of Multi-Rail-Aware Hierarchical MPI Reduce-Scatter and Allgather Operations". In: *ISC HIGH PERFORMANCE 2026*. ISC 2026. June 2026.

[6] **Chen, C.-C.**, Contini, N., Xu, L., Queiser, J., Subramoni, H., Panda, D. K., "Design and Implementation of Casting Compression for GPU-Aware MPI Collectives". In: *40th IEEE International Parallel & Distributed Processing Symposium*. IPDPS 2026. May 2026.

[7] Kuncham, G. K. R., Zhang, S., Mohammad, S., **Chen, C.-C.**, Panda, D. K., "MPI Communication Performance on AMD MI300A: Microbenchmarks and Applications". In: *Seventh Annual Workshop on Emerging Parallel and Distributed Runtime Systems and Middleware*. SC-W '25. Nov. 2025.

[8] **Chen, C.-C.**, Yao, J., Subramoni, H., Panda, D. K., "Design and Optimization of GPU-Aware MPI Allreduce Using Direct Sendrecv Communication". In: *54th International Conference on Parallel Processing*. ICPP 2025. Sept. 2025.

[9] **Chen, C.-C.**, Yao, J., Xu, L., Subramoni, H., Panda, D. K., "Unified Designs of Multi-rail-aware MPI Allreduce and Alltoall Operations Across Diverse GPU and Interconnect Systems". In: *39th IEEE International Parallel & Distributed Processing Symposium*. IPDPS 2025. June 2025.

[10] **Chen, C.-C.**, Kuncham, G. K. R., Subramoni, H., Panda, D. K., "Design and Implementation of Kernel-based MPI Reduction Operations for Intel GPUs". In: *31st IEEE International Conference on High Performance Computing, Data, and Analytics*. HiPC 2024. Dec. 2024.

[11] **Chen, C.-C.**, Kuncham, G. K. R., Kousha, P., Subramoni, H., Panda, D. K., "Design and Implementation of an IPC-based Collective MPI Library for Intel GPUs". In: *Practice and Experience in Advanced Research Computing 2024: Human Powered Computing*. PEARC '24. July 2024.

[12] **Chen, C.-C.**, Shafie Khorassani, K., Kousha, P., Zhou, Q., Yao, J., Subramoni, H., Panda, D. K., "MPI-xCCL: A Portable MPI Library over Collective Communication Libraries for Various Accelerators". In: *6th International Workshop on Emerging Parallel Distributed Runtime Systems and Middleware*. SC-W '23. Nov. 2023.

[13] Khuvis, S., Tomko, K., Brozell, S. R., **Chen, C.-C.**, Subramoni, H., Panda, D. K., "Optimizing Amber for Device-to-Device GPU Communication". In: *Practice and Experience in Advanced Research Computing*. PEARC '23. July 2023.

[14] **Chen, C.-C.**, Khorassani, K. S., Kuncham, G. K. R., Vaidya, R., Abduljabbar, M., Shafi, A., Subramoni, H., Panda, D. K., "Implementing and Optimizing a GPU-aware MPI Library for Intel GPUs: Early Experiences". In: *2023 IEEE/ACM 23rd International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*. May 2023.

[15] Khorassani, K. S., **Chen, C.-C.**, Subramoni, H., Panda, D. K., "Designing and Optimizing GPU-aware Nonblocking MPI Neighborhood Collective Communication for PETSc*". In: *2023 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*. May 2023.

[16] Suresh, K. K., Khorassani, K. S., **Chen, C.-C.**, Ramesh, B., Abduljabbar, M., Shafi, A., Subramoni, H., Panda, D. K., "Network Assisted Non-Contiguous Transfers for GPU-Aware MPI Libraries". In: *2022 IEEE Symposium on High-Performance Interconnects (HOTI)*. Aug. 2022.

[17] Shafie Khorassani, K., **Chen, C.-C.**, Ramesh, B., Shafi, A., Subramoni, H., Panda, D., "High Performance MPI over the Slingshot Interconnect: Early Experiences". In: *Practice and Experience in Advanced Research Computing*. PEARC '22. July 2022.

[18] **Chen, C.-C.**, Khorassani, K. S., Anthony, Q. G., Shafi, A., Subramoni, H., Panda, D. K., "Highly Efficient Alltoall and Alltoallv Communication Algorithms for GPU Systems". In: *2022 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. May 2022.

[19] Suresh, K. K., Ramesh, B., **Chen, C.-C.**, Ghazimirsaeed, S. M., Bayatpour, M., Shafi, A., Subramoni, H., Panda, D. K., "Layout-aware Hardware-assisted Designs for Derived Data Types in MPI". In: *2021 IEEE 28th International Conference on High Performance Computing, Data, and Analytics (HiPC)*. Dec. 2021.

[20] Shafie Khorassani, K., Hashmi, J., Chu, C.-H., **Chen, C.-C.**, Subramoni, H., Panda, D. K., "Designing a ROCm-Aware MPI Library for AMD GPUs: Early Experiences". In: *ISC High Performance 2021*. June 2021.

[21] **Chen, C.-C.**, Hasio, Y.-T., Lin, C.-Y., Lu, S., Lu, H.-T., Chou, J., "Using Deep Learning to Predict and Optimize Hadoop Data Analytic Service in a Cloud Platform". In: *2017 IEEE 15th Intl Conf on Dependable, Autonomic and Secure Computing, 15th Intl Conf on Pervasive Intelligence and Computing, 3rd Intl Conf on Big Data Intelligence and Computing and Cyber Science and Technology Congress(DASC/PiCom/DataCom/CyberSciTech)*. Nov. 2017.

Posters

[22] **Chen, C.-C.**, Yao, J., Subramoni, H., Panda, D. K., "Design and Optimization of GPU-Aware MPI Allreduce Using Direct Sendrecv Communication". In: *NVIDIA GTC AI Conference 2026*. May 2026.

[23] **Chen, C.-C.** "Designing GPU-Aware Collective Communication for Heterogeneous Clusters with Diverse GPUs and Interconnects". In: *Doctoral Showcase at The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC25)*. Nov. 2025.

[24] **Chen, C.-C.**, Yao, J., Xu, L., Subramoni, H., Panda, D. K., "Unified Designs of Multi-rail-aware MPI Allreduce and Alltoall Operations Across Diverse GPU and Interconnect Systems". In: *13th Annual MVAPICH User Group (MUG) Conference*. Aug. 2025.

[25] **Chen, C.-C.**, Kuncham, G. K. R., Subramoni, H., Panda, D. K., "Design and Implementation of a GPU-Aware MPI Collective Library for Intel GPUs". In: *ISC HIGH PERFORMANCE 2025*. June 2025.

[26] **Chen, C.-C.**, Xu, L., Subramoni, H., Panda, D. K., "Design and Implementation of MPI Collective Operations for Large Message Communication on AMD GPUs". In: *ISC HIGH PERFORMANCE 2025*. June 2025.

[27] **Chen, C.-C.**, Yao, J., Xu, L., Subramoni, H., Panda, D. K., "Unified Designs of Multi-rail-aware MPI Allreduce and Alltoall Operations Across Diverse GPU and Interconnect Systems". In: *39th IEEE International Parallel & Distributed Processing Symposium*. June 2025.

[28] **Chen, C.-C.**, Khorassani, K. S., Anthony, Q. G., Shafi, A., Subramoni, H., Panda, D. K., "MPI-xCCL: A Portable MPI Library over Collective Communication Libraries for Various Accelerators". In: *11th Annual MVAPICH User Group (MUG) Conference*. Aug. 2023.

[29] **Chen, C.-C.**, Khorassani, K. S., Anthony, Q. G., Shafi, A., Subramoni, H., Panda, D. K., "Highly Efficient Alltoall and Alltoallv Communication Algorithms for GPU Systems". In: *NVIDIA GTC23*. May 2023.